

Question	Marking Guidance	Mark	Comments
07.1	1. Vaccine/it contains antigen (from HPV); 2. Displayed on antigen-presenting cells; 3. Specific <u>helper T cell</u> (detects antigen and) stimulates specific B cell; 4. B cell divides/goes through mitosis/forms clone to give <u>plasma cells</u>; 5. B cell/plasma cell produces antibody;	4 max	1. Term 'antigen' may be first mentioned with point 2 2. Accept named example, e.g. macrophage/phagocyte/B cells 3. Accept 'helper T cell with receptor on surface' for 'specific' and B cells with receptor/antibody on surface that bind to antigen for 'specific'
07.2	1. Two (doses) because got more antibody; 2. With three doses, second dose/dose at 1 month doesn't lead to production of any more antibody (than the two-dose group)/get same/similar response; 3. Three doses would be more expensive/less popular with parents/girls (and serves no purpose);	2 max	1. Accept more effective in producing antibody 3. Accept 'less painful'
07.3	t-test, because comparing two means;	1	Mark for correct test <u>and</u> explanation correct Accept 'comparing the mean' Reject 'to show that the results/means are significant'

07.4	1. Compare (base sequences of) DNA; 2. Look for mutations/named mutations (that change the base sequence); 3. Compare (base sequences of) (m)RNA;	2 max	1 and 3 accept triplet/codon sequences for comparisons Ignore references to 'introns/non-coding DNA'
------	---	-------	---